Risk Analysis and Testing Plan

1- Risk Analysis

The Business Financial Monitoring System is a Python-based application designed to help businesses monitor their financial activities such as income, expenses, and profits. Since the system manages financial data and user interactions, it is important to analyze potential risks that may affect the system’s performance, reliability, and security.

Risk analysis helps identify possible problems, evaluate their impact, and define strategies to prevent or reduce these risks.

2- Risk Identification

Several risks may occur during the development or usage of the system

1. Data Loss *Financial data may be lost if the system crashes or if the JSON data file becomes corrupted.*	*3. User Input Errors* *Users may enter incorrect values such as negative numbers or wrong categories.*	*5. System Crash or Software Bugs* *Errors in the code or unexpected inputs may cause the application to stop working.*
2. Incorrect OCR Detection The OCR system used to read bills may detect incorrect numbers from images.	4. Unauthorized AccessIf authentication is weak, unauthorized users may access financial information.	6. File Export Errors Problems may occur when exporting reports to CSV or Excel files.

3- Risk Assessment and Management

he following table evaluates the probability and impact of each identified risk.

Risk level	*impact*	*probability*	*Risk*
High	High	Medium	Data loss
Medium	Medium	Medium	Incorrect OCR Detection
High	Medium	High	User input Error
Medium	High	Low	Unathorizaed Access
High	High	Medium	System Crash
Low	Medium	Low	File Export Errors

High-risk issues require stronger prevention strategies.

4. Risk Mitigation / Preventive Actions

To reduce the impact of risks, several mitigation strategies are implemented.

Mitigation Strategy	Risk
Automatically save data to a JSON file and create backups	Data Loss
Allow manual verification of detected bill amounts.	İncorrect OCR Detection
Implement input validation and category checks	User İnput Errors
Use a login and registration system with authentication	Unauthorized Access
Use error handling and exception management.	System Crash
Validate file paths and ensure proper file creation.	File Export Errors

These preventive actions help improve the reliability and safety of the system

5- Risk Matrix

The risk matrix visually represents the relationship between probability and impact

High	Medium	Low	Impact/Probability
Medium Risk	Low Risk	Low Risk	Low Impact
High Risk	Medium Risk	Low Risk	Medium Impact
Critical Risk	High Risk	Medium Risk	High Impact

In the risk matrix, probability is represented by the horizontal columns, and impact is represented by the vertical rows. The intersection between them determines the level of risk.

For example:

- Data Loss → High Impact + Medium Probability → High Risk
- User Input Errors → Medium Impact + High Probability → High Risk
- Export Errors → Medium Impact + Low Probability → Low Risk

This matrix helps prioritize which risks must be addressed first

6- Validation and Testing Plan

- **To ensure the system works correctly, several testing and validation methods will be used.**

6.1 Unit Testing

Each function in the program will be tested individually.

**add_income()*

**net _profit()*

**export_excel()*

*upload_bill()

*add_expense

*export_csv()

*search_by_data()

* add business()

These tests verify that each function performs the expected calculations.

6.2 Integration Testing

Integration testing ensures that different components of the system work together correctly.

Examples:

- **Adding income and expense updates the dashboard.**
- **Exporting reports includes the correct financial data.**

- OCR bill upload correctly adds expenses to the system.

6.3 System Testing

The entire application will be tested as a complete system.

Tests include:

Creating a business	**Adding financial transactions**
Exporting files	**Generating charts and reports**
Searching transactions by date	

6.4 Validation Methods

Validation ensures that the results produced by the system are accurate.

Validation will be performed by:

- **Comparing calculated profits with manual calculations.**
- **Testing the system using sample financial data.**
- **Verifying that exported reports match the dashboard data**

6.5 Performance Evaluation

The system performance will be evaluated based on:

Accuracy – Correct calculation of income, expenses, and profit	**Efficiency – The system should respond quickly to user .actions**	**Usability – The interface should be easy for users to understand.**
Reliability – The system should run without crashes	*****	*****

7-Success Criteria

The project will be considered successful if the following conditions are met:

The system correctly .tracks income and expenses	**Financial reports (CSV and Excel) are generated .successfully**	**The OCR feature detects .bill amounts from images**
Users can easily add businesses and transactions	Charts correctly visualize .financial information	The system stores and loads financial data without loss
The application runs without major errors or crashes	******	******

***The Business Financial Monitoring System has been tested and successfully meets all the specified requirements. It provides accurate financial information, generates reports correctly, and handles income and expenses efficiently. Any future errors or issues that may arise will be addressed through proper maintenance and updates**

Thank *you!*